

# Problem 7

**Problem 1** Determine if the function is even, odd, or neither.

(a)  $h(x) = x^4 + x^2 - 3$

**Explanation.** A function is even if  $f(x) = f(-x)$ , for all  $x$  in the domain, which means its graph is symmetric about the y-axis. A function is odd if  $f(-x) = -f(x)$ , for all  $x$  in the domain, which means its graph is symmetric about the origin.

$$h(x) = x^4 + x^2 - 3$$

$$h(-x) = (-x)^4 + (-x)^2 - 3 = x^4 + x^2 - 3$$

$h(x) = h(-x)$ . Hence  $h$  is even. This can be verified by graphing  $h$  and seeing that its graph is symmetric about the y-axis.

(b)  $s(t) = t^2 - t$

**Explanation.**

$$s(t) = t^2 - t$$

$$s(-t) = (-t)^2 - (-t) = t^2 + t$$

This does not equal  $s(t)$  so  $s(t)$  is not even.

$$-s(t) = -(t^2 - t) = -t^2 + t$$

This does not equal  $s(-t)$  so  $s$  is not odd. Hence,  $s(t)$  is neither even, nor odd.

(c) We know that  $\sin(\theta)$  is odd and  $\cos(\theta)$  is even. Is  $g(\theta) = \tan(\theta)$  even, odd, or neither?

**Explanation.**

$$g(\theta) = \tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$$

$$\begin{aligned} g(-\theta) &= \frac{\sin(-\theta)}{\cos(-\theta)} \\ &= \frac{-\sin(\theta)}{\cos(\theta)} \\ &= -\frac{\sin(\theta)}{\cos(\theta)} \end{aligned}$$

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$$-g(\theta) = -\frac{\sin(\theta)}{\cos(\theta)}$$

$$g(-\theta) = -g(\theta) \implies g \text{ odd}$$